

Station Black Out, Safety Analysis

March 3, 2026

1 Objective

You have been requested by the owners of the Zion nuclear power plant to analyze the possible consequences of a Station Black Out (SBO) situation with aggravated conditions and come up with a new system that allows to withstand the accidental situation for 72 hours. This exercise will be divided in several parts and deliveries. Each part will have each own delivery but it will be a unique report that will be growing through the course to finally become a final report. The following are the parts to be performed and delivered:

1. Steady state and SBO base case evaluation.
2. Development of an event tree for the SBO scenario.
3. Design of the new safety system.
 - (a) Deterministic safety analysis of the performance of the new safety system.
 - (b) Implications of the new system in the PSA event tree.

In Table 1 you can find a time table of sessions and deliveries.

Item	type	Date
Kick-off	class	03/03/2026
Base case calculation	class	27/03/2026
Part 1	delivery	07/04/2026
Part 2	tutoring	April 2026
Design of the new safety system (1)	class	8/05/2026
Design of the new safety system (2)	tutoring	May 2026
Defenses	class	27/05/2026
Final delivery	delivery	07/06/2026

Table 1: Sessions, deliveries and other important dates for the Team project

2 Part 1: base case calculation

2.1 RELAP model

In order to perform the exercise, a RELAP5 model of the NPP has been delivered with the layout displayed in Figure 1. This model has two different loops instead of 4; the first one is the broken loop and has the dimensions of a single loop, the other one is the intact loop and has the dimensions of three loops (three loops are lumped together in one single loop).

2.2 Steady state

The objective of this first simulation is to get acquainted with the Zion NPP model. This will be done by performing a steady-state calculation, where the plant conditions are stabilized. In order to set up a steady state, we need to adjust and control the following variables of our nodalization:

- Primary pressure
- Pressurizer level
- Primary mass flow, pump speed
- Core power
- Feedwater flow, secondary pressure and SG secondary side water level

The adjustments have already been done; you will only need to assure that everything works well. Table 2 shows the steady state values at nominal operation of the Zion NPP, these are our target values.

Parameter	Expected value
Power (MW)	3250
Primary pressure in the hot leg (MPa)	15.5
Secondary pressure (MPa)	6.7
Pressurizer level (m)	8.8
Broken Steam Generators DC level (m)	12.2
Intact Steam Generators DC level (m)	12.2
Core outlet temperature (K)	603
Core inlet temperature (K)	571
Broken Main feedwater flow (kg/s)	459
Intact Main feedwater flow (kg/s)	1325
Intact loop flow (kg/s)	12865
Broken loop flow (kg/s)	4351

Table 2: Steady state conditions

To run the case execute relap in command line with:

- **First!!!** Make a copy of restart.r to yourfile.r. In this way you keep a copy of the original restart.r
- **cp restart.r yourfile.r**
- **relap.sh yourfile restart**

With the calculation results, obtain the values for the parameters of Table 2 and compare with your results. In addition, make plots of some of the parameters to evaluate if the parameters are steady or not.

2.3 SBO: Base case

We will consider that SBO takes place at 0 seconds with aggravated conditions. It implies that all systems that require power will be unavailable. This will lead to the trip of the reactor coolant pumps, the fall of the control rods and cease of other important systems like the cooling of the pump seals. The general conditions that will be common for all sequences are as follows:

- SBO at 0.0 (in your input it will be at 1000.0)
- SCRAM
- trip of RCPs
- trip of FW and AFW
- trip of PZR heaters and volumetrical control system
- Unavailability of HPSI and LPSI throughout the transient
- Main steam line valves fully closed with scram signal
- Safety/relief valves actuate according to high SG pressure set points

In this project, we want to evaluate all possible sequences in order to quantify the possible risk reduction thanks to our new safety system. In this sense we need to configure the input so we can adjust the different possible sequences. The possible actions are listed below:

- The seal of the pumps will not be cooled, so we may have a small leak at the pump location starting from 10 minutes after SBO initiation. The leak size will be 0.00001 m² per pump (remember that the intact loop represents 3 loops). The leaks are already implemented, locate them in the input and make sure that they open at the right time and with the right area.
- According to NRC NUREG/CR-7110 vol 2. the leak may increase when saturation takes place at the RCP. The area will increase 4x. Duplicate the valve/tdpv components for the small leak and open these new ones when the temperature at the pumps reaches saturation conditions.
- Auxiliary feed water by means of a turbo-driven pump (TDP) which is controlled manually (requires batteries).
- Batteries that allow for prolonged control of the TDP lasts for 5h. The TDP becomes uncontrolled after 5h, this means that water is injected to the SG until the separator is flooded at which point the TDP stops. [check out the file TDPinstructions.i for further support](#)
- According to the EOPs, operators should follow a cooldown strategy. This is performed by cooling down the SG at a rate of 55 K/h until 15 bar. [check out the file cooldown.i for further support](#)
- According to the EOPs, operators should depressurize the primary system if the Core Exit Temperature is higher than 923 K (use vapor temperature to represent core exit thermocouples). This will be performed by fully opening the Pressurizer relief valves.
- Recovery of AC power after 12h. This means that you recover AFW, HPIS and LPIS at this time.

The tasks you will need to perform in this part of the project are:

1. Configure and run the base case. You should consider that all the above actions/options take place.
2. Verify that all the conditions are fulfilled.
3. Describe the evolution of the transient and the different phenomena that take place.
4. Make two additional calculations where you don't follow one of the two EOPs actions (avoid one for each additional calculation). Compare the three cases and discuss the EOPs.

2.4 Delivery

You should deliver the first draft of the final report with the draft contents of this part. Keep in mind the general structure provided in Appendix A. The purpose of this delivery is only to track your progress through the course. This delivery will be graded with a binary mark (passed or not passed). Make sure to complete each task in time, otherwise it will be difficult to complete the project satisfactorily.

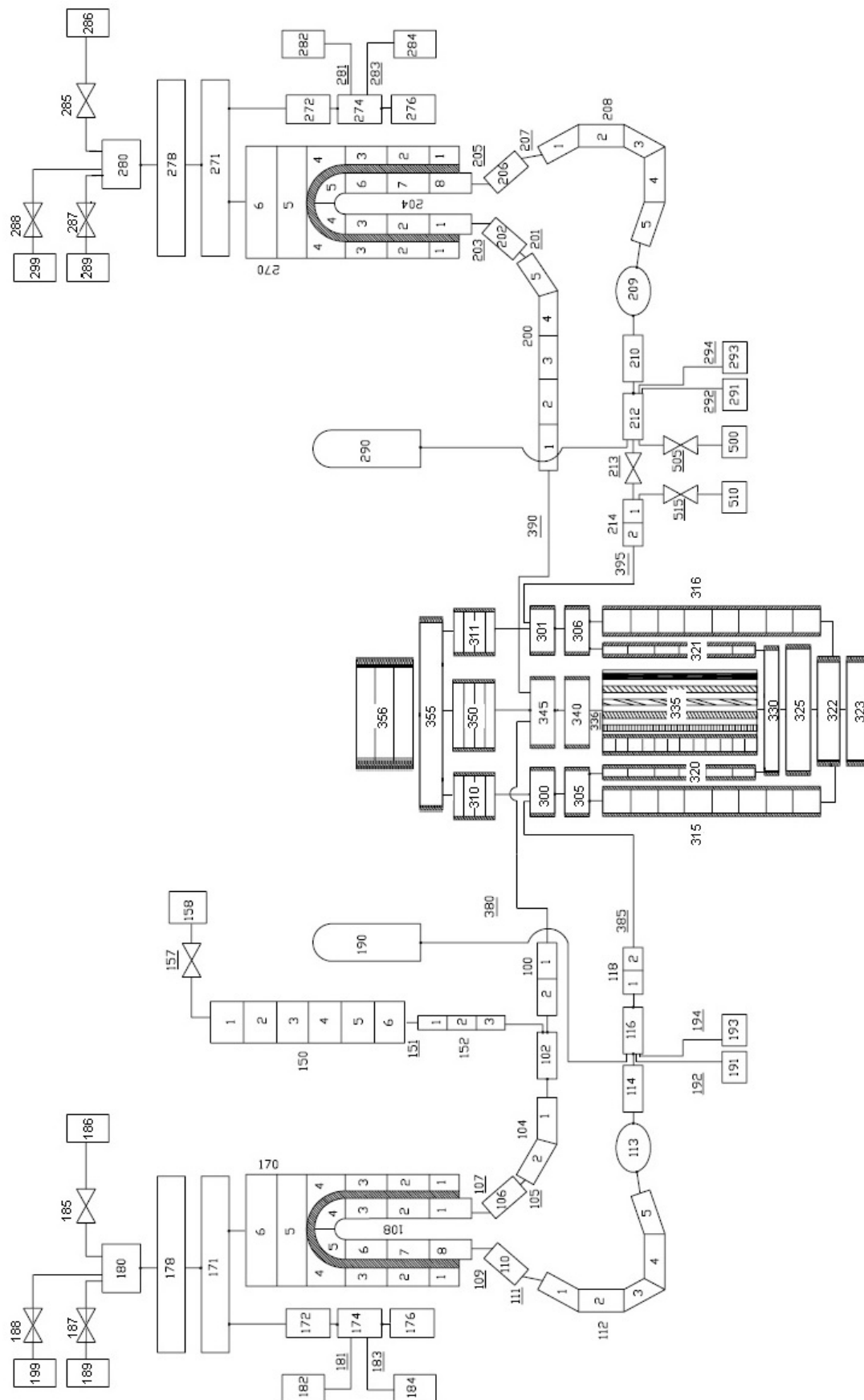


Figure 1: RELAP5 nodalization of the Zion NPP

3 Part 2: Event Tree

Even though, most of the systems are unavailable, there are still a few systems/actions prone to be part of an event tree. In this part of the assignment you will create an event tree of all the possible sequences that may happen in the SBO conditions. Here are the options available:

1. The seal of the pumps will not be cooled, so we may have a small leak at the pump location starting from 10 minutes after SBO initiation. The leak size will be 0.00001 m² per pump (remember that the intact loop represents 3 loops)(available, not available).
2. According to NRC NUREG/CR-7110 vol 2. the leak may increase when saturation takes place at the RCP.
3. Auxiliary feed water by means of a turbo-driven pump (TDP) (available, not available).
4. Batteries that allow for prolonged control of the TDP. Otherwise, the TDP becomes uncontrolled after 5h, this means that water is injected to the SG until the separator is flooded at which point the TDP stops. (available, not available).
5. Recovery of AC power after 12h. (available, not available)

3.1 Configure and run all possible sequences

After you have defined your event tree, you will need to configure a matrix of cases, evaluate which ones will present core damage (CD) and success, then run the sequences. Compare the outcome of the calculations with your pre-evaluations. In addition to writing CD in the consequence column of your event tree, write the time of CD.

3.2 PSA calculations

For each of the sequences assign a frequency and calculate the total CD frequency.

3.3 Delivery

You should deliver the second draft of the final report with the draft contents of the first and second parts. Keep in mind the general structure provided in Appendix A. This delivery will be graded with a binary mark (passed or not passed). Make sure to complete each task in time, otherwise it will be difficult to complete the project satisfactorily.

4 Part 3: New Safety System

In this last part of the assignment, your team needs to design and implement a passive safety system that allows to cope with the SBO scenario for at least 72 hours for any of the postulated sequences (with RCP cooling). We will focus on the first conceptual design and testing, this means that at this stage we will not consider standards, single failure criteria and operability of the system.

Start this part by reading carefully sections 1 and 2 of ref. [1]. Discuss with your group which of the systems you think will be better for coping with the SBO scenario. Come to the next class with a clear idea and a sketch of the nodalization to be implemented in RELAP. Input materials will be provided to ease the implementation. The steps to be performed in this part of the team project are:

- Initial conceptual design of your safety system.
- Dimensioning of the pipes, pools and volumes of water. By doing hand calculations you should be able to roughly estimate the design geometrical characteristics and the working boundary conditions.
- Submit the design to the RELAP development team (ANT research group), they will provide an initial input deck that you can implement into your Zion model.
- Select the worse case among the cases **without** RCP leak and simulate the scenario to proof by a deterministic calculation that the power plant could withstand the accident for 72 hours with the new system.
- Discuss the implications of the implementation of the new system and write a recommendation to the owners of the Zion nuclear plant.

4.1 Delivery

Delivery 1: The final report of the project as detailed in Appendix A. **Delivery 2:** The input file for each relevant performed calculation.

References

- [1] IAEA. Passive Safety Systems and Natural Circulation in Water Cooled Nuclear Power Plants. Technical report, IAEA, 2009.

A Structure of the final report

The structure of the final report should be similar (just a general idea) to the following:

1. Introduction
 - (a) Objective
 - (b) Description of the scenario
 - (c) Safety implications of the scenario
 - (d) Brief description of the work and final conclusion
2. Methodology
 - (a) Model and codes
3. Analysis of the base case
 - (a) Description of the BC
 - (b) Description of the results
 - (c) Possible EOP actions
4. Event tree and results
 - (a) Description of the event tree generation
 - (b) PSA analysis
 - (c) Description of the overall results
5. New safety system
 - (a) Description
 - (b) New design implementation for safety analysis
 - (c) performance of the new design (simulation of the worst case, DSA)
 - (d) performance of the new design for risk (PSA)
6. Conclusions
7. Author credit statement¹

¹<https://www.elsevier.com/researcher/author/policies-and-guidelines/credit-author-statement>